

Dharma Ransomware

Malware Family Documentation Report

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Topic: Cybersecurity & Malware Analysis

Introduction :-

Dharma Ransomware is a well-known ransomware family targeting Windows-based systems and enterprise environments. It commonly exploits exposed Remote Desktop Protocol (RDP) services and weak passwords to gain unauthorized access. Once executed, the malware encrypts files and demands ransom payments in cryptocurrency.

1. Family Name and Aliases:-

Family Name:

Dharma Ransomware is a well-known ransomware family that targets Windows-based systems and enterprise networks. It is mainly used for financial extortion by encrypting files and demanding ransom payments in cryptocurrency.

Aliases:

Dharma ransomware is also known by several other names, including:

- CrySiS
- Dharma-CrySiS
- Dharma Variant Family

These aliases exist because Dharma shares similarities with the older CrySiS ransomware codebase and evolved from it over time.

Category:

Type: Ransomware

Subcategory: Ransomware-as-a-Service (RaaS)

2. Discovery History and Timeline :-

Dharma Ransomware is a highly active ransomware family that emerged in the mid-2010s and became one of the most widespread ransomware threats targeting businesses and enterprise environments. It gained attention due to its aggressive use of exposed Remote Desktop Protocol (RDP) services and its continuously evolving variants.

3. Original Purpose and Evolution :-

Original Purpose :-

The primary purpose of Dharma Ransomware was financial extortion. It was designed to:

1. Encrypt victim files and systems
2. Prevent access to important data
3. Demand ransom payments in cryptocurrency for file recovery

The malware mainly targeted:

- Windows-based systems
- Small and medium businesses
- Enterprise networks
- Attackers used Dharma to generate profit by forcing victims to pay for decryption keys.

Evolution of Dharma Ransomware:-

2016: Dharma started as a basic ransomware focused on encrypting files and demanding ransom payments.

2017–2018: It evolved to target businesses and enterprise networks through exposed RDP services.

2019–2020: Attackers added advanced techniques like credential theft, PowerShell scripts, and lateral movement.

2020 onwards: Dharma adopted the Ransomware-as-a-Service (RaaS) model, allowing affiliates to conduct attacks.

Recent versions: Some campaigns use double extortion, where attackers steal sensitive data before encryption and threaten to leak it publicly.

4. Technical Characteristics and Behavior :-

Technical Characteristics:-

Uses AES and RSA encryption to lock files.

Encrypts documents, images, databases, backups, and network-shared files.

- Adds custom extensions like:
- .dharma
- .wallet
- .cobra

Creates ransom notes such as:

1. FILES ENCRYPTED.txt
2. Info.hta

Behavior:-

Commonly spreads through exposed RDP services and weak passwords.

Disables antivirus and backup services.

Deletes Windows shadow copies to prevent recovery.

Maintains persistence using registry keys and scheduled tasks.

Some variants steal credentials and move across networks.

5. Common File Extensions and Packaging Formats :-

Common File Extensions:-

After encrypting files, Dharma ransomware adds custom extensions such as:

.dharma

.wallet

.cobra

.java

.arrow

Packaging Formats:-

Dharma ransomware is commonly distributed as:

EXE files (.exe)

DLL loaders (.dll)

Self-extracting archives

Packed or obfuscated binaries

6. Known Hashes and Detection Signatures :-

Known Hashes:-

Security researchers use hashes to identify Dharma ransomware samples.

Detection Signatures:-

Common indicators used to detect Dharma ransomware include:

- Creation of ransom notes:
 - FILES ENCRYPTED.txt
 - info.hta
 - File extension changes such as:
 - .dharma
 - .wallet
 - Deletion of Windows shadow copies

7. Associated Threat Actors and Campaigns :-

Associated Threat Actors:-

- Dharma ransomware is mainly linked to:
- Financially motivated cybercriminal groups
- Ransomware affiliates
- RDP brute-force attackers
- Initial access brokers

Unlike some ransomware families, Dharma is not strongly connected to a single nation-state APT group.

Common Campaigns:-

Dharma campaigns commonly target:

- Small and medium businesses (SMBs)
- Healthcare organizations
- Educational institutions
- Enterprise networks

Common Attack Methods :-

- Exploiting exposed Remote Desktop Protocol (RDP) services
- Weak or stolen credentials
- Manual ransomware deployment after network access

8. Typical TTPs and Infection Chain :-

Typical TTPs (Tactics, Techniques, and Procedures):-

Initial Access :-

- Exploits exposed RDP services
- Uses weak or stolen credentials
- Performs brute-force login attacks

Execution:-

- Executes ransomware manually after gaining access
- Uses PowerShell and command-line tools

Persistence :-

- Creates registry Run keys
- Uses scheduled tasks and startup folders

Defense Evasion :-

- Disables antivirus software
- Deletes shadow copies and backups
- Credential Access

Uses tools like:-

- Mimikatz
- PowerShell scripts

Lateral Movement:-

- Spreads through network shares
- Accesses other systems using stolen credentials

Infection Chain:-

- Scan for exposed RDP ports
- Brute-force or steal credentials
- Gain administrator access
- Upload Dharma ransomware payload
- Disable security and backup services
- Encrypt files and shared drives
- Display ransom notes and demand payment

9. Network Indicators :-

Common Network Indicators:-

- Dharma ransomware infections often involve:
- Unusual RDP traffic spikes
- Connections to attacker-controlled servers
- Suspicious outbound network activity
- TOR-related communication

Detection Signs :-

- Security teams may detect:
- High RDP authentication failures
- Unexpected outbound traffic
- Encrypted traffic to suspicious servers
- Sudden file-sharing activity across networks

10. Persistence and Exfiltration Behavior :-

Persistence:-

- Dharma ransomware maintains access by:
- Creating registry Run keys
- Using startup folders
- Scheduling tasks to run after reboot

Exfiltration Behavior:-

- Some modern Dharma variants:
- Steal sensitive data before encryption
- Collect credentials and documents
- Use double extortion by threatening to leak stolen data publicly

11. File System Artifacts :-

Common File Artifacts:-

Dharma ransomware creates files such as:

- FILES ENCRYPTED.txt
- Info.hta

These files contain ransom instructions and attacker contact details.

Encrypted File Pattern:-

After encryption, files are renamed with:

- Victim ID
- Attacker email
- Custom extension

12. Defensive Recommendations :-

Preventive Measures :-

- Disable exposed RDP services if not required
- Enable Multi-Factor Authentication (MFA)
- Use strong and unique passwords
- Keep systems and software updated
- Maintain secure offline backups

Detection and Monitoring:-

- Monitor suspicious RDP login attempts
- Detect unusual PowerShell activity
- Watch for sudden file extension changes

- Monitor shadow copy deletion commands

Security Best Practices :-

- Use endpoint protection and antivirus tools
- Apply network segmentation
- Restrict administrator privileges
- Conduct regular security awareness training

Conclusion:-

Dharma ransomware continues to remain a significant cybersecurity threat. Organizations should implement strong security measures, monitor remote access systems, and maintain secure backups to reduce the risk of attacks.

References :-

- MITRE ATT&CK
- CISA Malware Reports
- VirusTotal
- Malpedia